1. Cooling system and how a pressure sensor is used

Basic Physics of Cooling Battery and Pressure:
Bernoulli's Principle: Total energy = Static Pressure + density*velocity^2
(If the kinetic energy of the air increases, the potential energy of the static pressure must drop)

PV = nRT

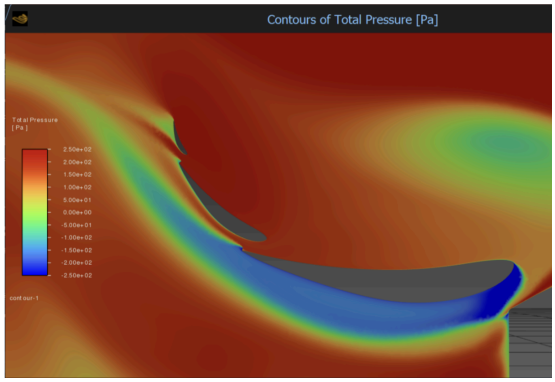
Assuming an air-cooled accumulation system: an accumulator has a sensor to measure the volumetric pressure drop

- We need one sensor at the inlet, and another sensor at the outlet → **differential pressure sensor**

2. Pressure sensor Specifications that are important for this application:

1. **Range**: the smallest to largest value the sensor is capable of measuring.

2. **Resolution**: the smallest change is that the sensor can distinguis/report
Ex. 10005 Pa vs 10004 Pa if the sensor has a resolution of 1 Pa
3. **Accuracy**
4. **Precision**: if I measure repeatedly, how often do I get the same result?



Based on this graph, the entire aerodynamic phenomenon I am trying to measure is just 500 Pa wide.

- Atmospheric pressure is 101,325 Pa

3. Different Types of Sensors

Absolute vs. differential pressure sensors:
Absolute:

Differential pressure sensor:
could be used for things like intake restrictor monitoring (you have Port A and Port B; do we bring these 2 ports together?)

- **MEMS**: its mechanical pressure diaphragm is down to the micrometer level. High pressure pushes on the silicon diaphragm, and lower air pressure pushes up against the bottom. P1 > P2 → the diaphragm bends down a few nanometers → ASIC passes current through resistors and measures the change in voltage bc of the bending → converts it to Pa through a built-in calibration equation.
    - Extremely lightweight, take up a few square milimeters, measures pressure changes down to 0.1 Pa, chassis vibrations don't fake out this sensor bc the moving mass of its diaphragm is so low.
    - It has an onboard digital ASIC which allows you to chain multiple sensors along the I2C

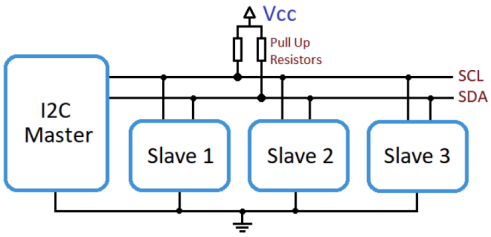
4. Communication Interfaces for Sensors

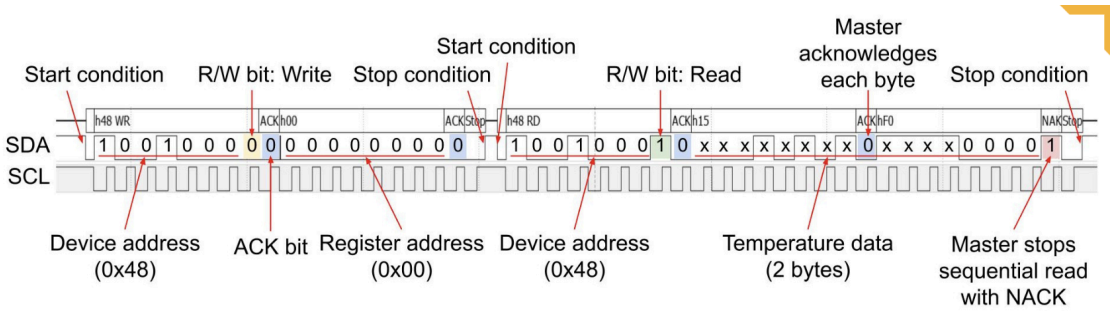
Types of Communication interfaces and trade-offs

Type of Interface	Data rate	Distance	Power / Pins	Notes
SPI	25 Mbps high	~20cm, Short best for comm on the same PCB or nearby boards	4 shared signal lines (SCLK, MOSI, MISO)	fast and good for multiple identical sensors because each can have its own pin.

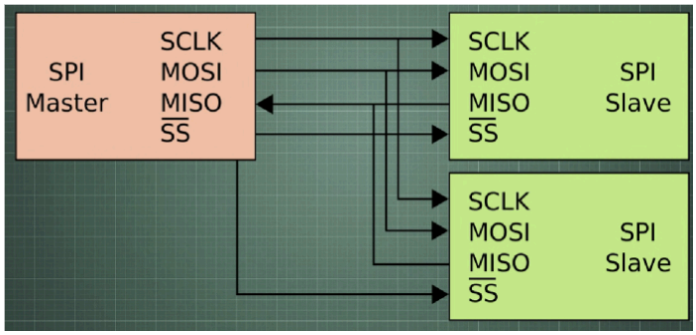
I2C	1Mbps, medium	~1m, same PCB or short wiring	2 shared signal pins(SDA and SCL)	many devices can share the same bus, but every device needs a unique address.
Onewire	low	longer wiring than SPI/I2C, depd on layout	Usually 1 data pin plus ground	each device has a unique ID, so many devices can share *one wire* Not used for high-speed sensors.
Analog	no fixed digital data rate	Short to moderate, but longer wires are susceptible to noise	Usually 1 analog input per sensor	sensor outputs a voltage/current requires an ADC
UART	25kbps, low	~15m, longer than PCB buses if designed carefully	2 signal pins (TX and RX)	1 transmitter talks to 1 receiver →not designed for identical sensors sharing 1 bus

I2C:

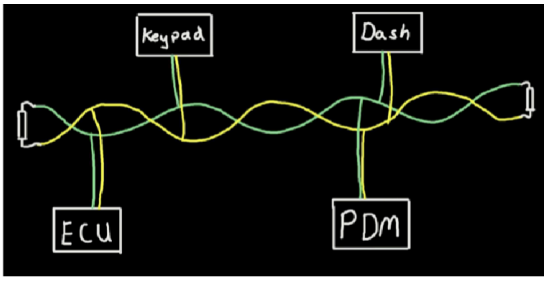




SPI:



CAN:



PCB communication through different protocols like SPI, I2C. Physically - multiple layers in PCB to route signals (and power)

- intra-PCB communication happens in the layers of the board:

